

Assignment No 1



- Title: Development of Tic Tac Toe Game program.
- Problem Statement:
Create a program to simulate a game of tic tac toe with following features.
 1. A 3×3 Board for gameplay.
 2. Two Players, one computer and one human represented by symbol 'O' and 'X'.
 3. Randomized moves to decide the sequence of actions.
 4. Automatic detection of win or draw conditions.

- Algorithm:

1: Initialize.

- Create a 3×3 Board Gameplay filled with empty space(" ")
- Create and shuffle an array representing possible moves.

2 - Show instructions

- Display the layout of board and numbered cells (1 to 9)

3 - Start Gameplay:

Repeat until:

A win condition is met.

All moves are exhausted.

on each turn

Use the move index to calculate the row and column for the current move.



1. Place the player's symbol ('O' for the computer and 'X' for human)
3. Display the updated board.
4. Increment the move index.
5. Switch the turn between players.

Inlin-draw check.

- After each move.
 - check if any row, column or a diagonal is completely occupied by the same symbol
- If so, declare the current turn's opponent as the winner.
- If all the cells are filled with no winner, declare the draw.

Declare Result:

- Output the winner or announce a draw

End.

- Exit the program.

Conclusion

Thus we have Successfully implemented tic-tac-toe using java.

public class File {

static String[] board;

static String turn;

static String checkWinner() {

for (int a = 0; a < 8; a++) {

String line = null;

switch (a) {

case 0: line = board[0] + board[1] + board[2]; break;

case 1: line = board[3] + board[4] + board[5]; break;

case 2: line = board[6] + board[7] + board[8]; break;

case 3: line = board[0] + board[3] + board[6]; break;

case 4: line = board[1] + board[4] + board[7]; break;

case 5: line = board[2] + board[5] + board[8]; break;

case 6: line = board[0] + board[4] + board[8]; break;

case 7: line = board[2] + board[4] + board[6]; break;

}

if (line.equals("XXX")) return "X";

else if (line.equals("OOO")) return "O";

}

for (String s : board) {

if (s.matches("[1-9]")) return null;

}

return "draw";

}

static void printBoard() {

System.out.println("|---|---|---|");

System.out.println("| " + board[0] + " | " + board[1] + " | " + board[2] + " |

");

System.out.println("|-----|");

System.out.println("| " + board[3] + " | " + board[4] + " | " + board[5] + " |

");

System.out.println("|-----|");

System.out.println("| " + board[6] + " | " + board[7] + " | " + board[8] + " |

");

System.out.println("|---|---|---|");

}

static void computerMove() {

Random rand = new Random();

int move;

do {

move = rand.nextInt(9);

} while (!board[move].matches("[1-9]"));

board[move] = "O";

System.out.println("Computer chose: " + (move + 1));

}


```

public static void main(String[] args) {
    Scanner in = new Scanner(System.in);
    board = new String[9];
    turn = "X";
    String winner = null;

    for (int a = 0; a < 9; a++) board[a] = String.valueOf(a + 1);

    System.out.println("Welcome to Tic-Tac-Toe!");
    printBoard();

    while (winner == null) {
        if (turn.equals("X")) {
            System.out.print("Your turn (1-9): ");
            int numInput;
            try {
                numInput = in.nextInt();
                if (! (numInput > 0 && numInput <= 9) || !board[numInput -
1].matches("[1-9]")) {
                    System.out.println("Invalid move, try again.");
                    continue;
                }
            } catch (InputMismatchException e) {
                System.out.println("Invalid input, try again.");
                in.next();
                continue;
            }
            board[numInput - 1] = "X";
        } else {
            computerMove();
        }
        printBoard();
        winner = checkWinner();
        turn = (turn.equals("X")) ? "O" : "X";
    }

    System.out.println(winner.equals("draw") ? "It's a draw!" : "Congratulations!
" + winner + " wins!");
    in.close();
}

```


Output :

```
PS C:\Users\lenovo\Desktop> cd "C:\Users\lenovo\Desktop" & if ($?) { java File.java } & .
(57) ( java File )
Welcome to Tic-Tac-Toe!
|---|---|---|
| 1 | 2 | 3 |
|---|---|---|
| 4 | 5 | 6 |
|---|---|---|
| 7 | 8 | 9 |
|---|---|---|
Your turn (1-9):
```

```
TERMINAL
| 4 | X | 6 |
|---|---|---|
| 7 | 8 | 9 |
|---|---|---|
Computer chose: 7
|---|---|---|
| 1 | 2 | 3 |
|---|---|---|
| 4 | X | 6 |
|---|---|---|
| 0 | 8 | 9 |
|---|---|---|
Your turn (1-9):
```

```
Computer chose: 9
|---|---|---|
| X | 2 | 3 |
|---|---|---|
| 4 | X | 6 |
|---|---|---|
| 0 | 8 | 0 |
|---|---|---|
Your turn (1-9):
```

```
Computer chose: 3
|---|---|---|
| X | 2 | 0 |
|---|---|---|
| X | X | 6 |
|---|---|---|
| 0 | 8 | 0 |
|---|---|---|
Your turn (1-9):
```



```
YOUR TURN (1-9): 0
```

```
|---|---|---|
```

```
| X | 2 | 0 |
```

```
|---|---|---|
```

```
| X | X | X |
```

```
|---|---|---|
```

```
| 0 | 9 | 0 |
```

```
|---|---|---|
```

```
Congratulations! X wins!
```

```
PS C:\Users\lenovo\Desktop>
```

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